NUCLEI

FORMULA SHEET | JEE MAINS | Let's Play With Physics

1. COMPOSITION & NOTATION

$$^{A}_{Z}X: \quad Z = \text{protons}, \quad A = \text{nucleons}$$
$$N = A - Z = \text{neutrons}$$

Term	Same	Example
Isotopes	Z	$^{1}_{1}$H, $^{2}_{1}$H, $^{3}_{1}$H
Isobars	A	$^{3}_{1}$H, $^{3}_{2}$He
Isotones	N	$^{3}_{1}$H, $^{4}_{2}$He
Mirror nuclei	$Z \leftrightarrow N$	$^{3}_{1}$H, $^{3}_{2}$He

Quantity	Value
1 u	$\frac{1}{12}m(^{12}\text{C}) = 1.6605 \times 10^{-27}$ kg
1 u $\cdot c^2$	931.5 MeV
m_p	1.007276 u
m_n	1.008665 u
m_H (atom)	1.007825 u
m_e	0.000549 u (0.511 MeV/c^2)
1 fm	10^{-15} m

⋆ Common Pitfall

- Neutron is slightly heavier than proton ($m_n > m_p$)
- Isotopes have identical chemistry, different nuclear properties

2. NUCLEAR SIZE & DENSITY

$$R = R_0 A^{1/3}, \qquad R_0 \approx 1.2 \text{ fm}$$
$$\frac{R_1}{R_2} = \left(\frac{A_1}{A_2}\right)^{1/3}, \qquad V \propto A$$
$$\rho = \frac{A\,m_u}{\frac{4}{3}\pi R_0^3 A} = \frac{3m_u}{4\pi R_0^3} \approx 2.3 \times 10^{17} \text{ kg/m}^3$$

⋆ Common Pitfall

- Nuclear density is the **same for all nuclei** (independent of A)
- Doubling the radius needs 8× the mass number

3. MASS DEFECT & BINDING ENERGY

$$E = mc^2, \qquad 1 \text{ u} \leftrightarrow 931.5 \text{ MeV}$$

Mass defect:

$$\Delta m = [Zm_p + (A-Z)m_n] - M_{\text{nucleus}}$$

Using atomic masses: $\Delta m = Zm_H + (A-Z)m_n - M_{\text{atom}}$

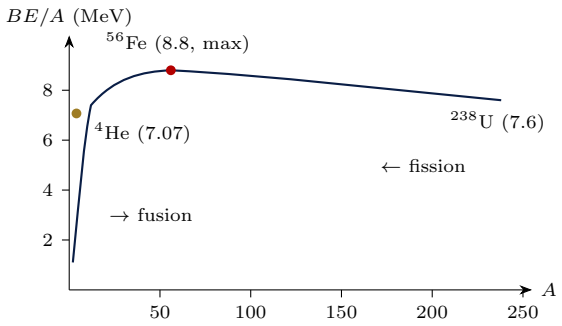
Binding energy: $BE = \Delta m\, c^2 = \Delta m(\text{u}) \times 931.5$ MeV

BE per nucleon: $\overline{B} = \dfrac{BE}{A}$

⋆ Common Pitfall

- With atomic masses, use m_H (not m_p) — electron masses then cancel
- Higher $BE/A \Rightarrow$ more stable nucleus (not higher total BE)
- Mass of a nucleus is always **less** than the sum of its free nucleons

4. BE/A vs MASS NUMBER

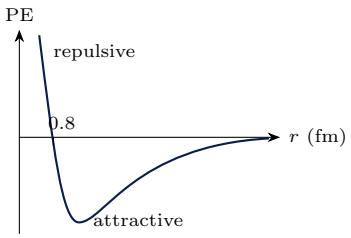


- Nearly constant (≈ 8 MeV) for $30 < A < 170 \Rightarrow$ **saturation** of nuclear force
- Low for light & heavy nuclei $\Rightarrow$ both **fusion** and **fission** release energy

Energy released:

$$Q = \sum BE_{\text{products}} - \sum BE_{\text{reactants}} = \sum(\overline{B}A)_f - \sum(\overline{B}A)_i$$

5. NUCLEAR FORCE



- Strongest force in nature; very **short range** ($\sim$ 2–3 fm)
- Attractive for $r > 0.8$ fm, strongly repulsive below
- **Charge independent**: n–n, p–p, n–p nearly equal
- **Saturating**: a nucleon interacts only with nearest neighbours
- Non-central, does not obey inverse-square law

6. NUCLEAR REACTIONS & Q-VALUE

Reaction: $a + X \rightarrow Y + b$

$$Q = (m_a + m_X - m_Y - m_b)\,c^2 = K_{\text{final}} - K_{\text{initial}}$$

$Q > 0$: exothermic (energy released); $Q < 0$: endothermic

Conserved: charge ($\sum Z$), nucleon number ($\sum A$), energy, momentum, angular momentum

⋆ Common Pitfall

- Mass is NOT conserved separately; mass–energy is
- Balance A and Z first to find the unknown particle

7. NUCLEAR FISSION

$$^{235}_{92}\mathrm{U} + {}^{1}_{0}n \rightarrow {}^{236}_{92}\mathrm{U}^{*} \rightarrow {}^{141}_{56}\mathrm{Ba} + {}^{92}_{36}\mathrm{Kr} + 3\,{}^{1}_{0}n + \sim 200 \text{ MeV}$$

Energy per fission $\approx$ 200 MeV $= 3.2 \times 10^{-11}$ J ($\approx$ 0.9 MeV/nucleon)

Fissions per second for power P: $n = \dfrac{P}{E_{\text{fission}}}$

Multiplication factor k: $k = 1$ critical, $k > 1$ supercritical, $k < 1$ subcritical

Reactor: moderator (heavy water, graphite) slows neutrons; control rods (Cd, B) absorb neutrons; coolant removes heat

⋆ Common Pitfall

- U-235 fission needs **slow** (thermal) neutrons
- Uncontrolled chain reaction ($k > 1$) $\rightarrow$ atom bomb; controlled ($k = 1$) $\rightarrow$ reactor

8. NUCLEAR FUSION

p–p cycle (Sun): $4\,{}^{1}_{1}\mathrm{H} \rightarrow {}^{4}_{2}\mathrm{He} + 2e^{+} + 2\nu + 26.7$ MeV

D–T: ${}^{2}_{1}\mathrm{H} + {}^{3}_{1}\mathrm{H} \rightarrow {}^{4}_{2}\mathrm{He} + {}^{1}_{0}n + 17.6$ MeV

Needs $T \sim 10^{7}$ K to overcome Coulomb repulsion (thermonuclear)

⋆ Common Pitfall

- Energy **per nucleon**: fusion ($\sim$ 6.7 MeV) > fission ($\sim$ 0.9 MeV)
- Energy **per reaction**: fission (200 MeV) > fusion

9. RADIOACTIVITY (QUICK REFERENCE)

$$N = N_0 e^{-\lambda t} = N_0 \left(\tfrac{1}{2}\right)^{t/T_{1/2}}$$

$$T_{1/2} = \frac{0.693}{\lambda}, \qquad \tau = \frac{1}{\lambda} = 1.44\,T_{1/2}$$

Activity: $A = \lambda N = A_0 e^{-\lambda t}$; 1 Ci $= 3.7 \times 10^{10}$ Bq

Decay	Change	Process
α	$A-4$, $Z-2$	emits ${}^{4}_{2}\mathrm{He}$
β^{-}	$Z+1$	$n \rightarrow p + e^{-} + \bar{\nu}$
β^{+}	$Z-1$	$p \rightarrow n + e^{+} + \nu$
γ	no change	excited nucleus de-excites

No. of decays: $n_{\alpha} = \dfrac{\Delta A}{4}$, $n_{\beta^{-}} = 2n_{\alpha} - \Delta Z$

10. JEE SHORTCUTS & MASTER TIPS

1. $R \propto A^{1/3}$; density same for all nuclei
2. $BE = \Delta m \times 931.5$ MeV (Δm in u)
3. $Q = \sum BE_f - \sum BE_i$ (use $\overline{B} \times A$)
4. Peak of BE/A at $A \approx 56$ (Fe), ≈ 8.8 MeV
5. Fission $\approx$ 200 MeV; fusion D–T = 17.6 MeV
6. $T_{1/2} = 0.693/\lambda$; $\tau = 1.44\,T_{1/2}$

⋆ Master Confusion List

1. **Stability:** decided by BE/A, not total BE
2. **Atomic vs nuclear mass:** use m_H with atomic masses
3. **Density** independent of A; **radius** $\propto A^{1/3}$
4. **Fission vs fusion:** per nucleon fusion wins; per event fission wins
5. Q **sign:** $Q > 0$ means products are lighter / more tightly bound
6. **Nuclear force** is charge independent; Coulomb force acts only on protons

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